EurJOC
European Journal of Organic Chemistry

Full Paper
doi.org/10.1002/ejoc.202000534

Chemistry Europe
European Chemical Societies Publishing

Chemoselective Reduction

Electrochemical Reduction of α-Amino Allenylphosphonates to α-Amino Allylphosphonates

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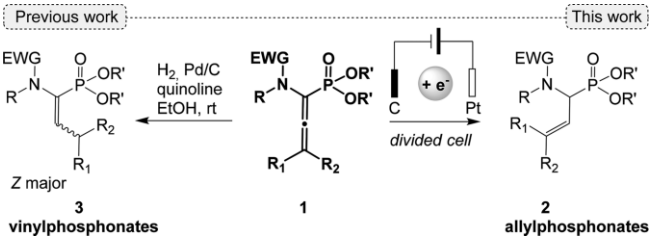
In memory of François Couty

Abstract: α-Amino allenylphosphonates are reduced to α-amino allylphosphonates using electrochemical processes. The reduction products are obtained in moderate to good yields. Moreover, in the case of tosylamine compounds, we were pleased to take advantage of a simultaneous reduction and cleavage of the tosyl group from the nitrogen atom to access unprotected amino-allylphosphonates. The mechanism of this reduction is also discussed.

Introduction

Due to their easiness of preparation through the Kabachnik–Fields reaction,[1,2] α-aminophosphonates have been extensively studied.[3] The latter exhibit a wide range of biological activities such as enzyme inhibition or antibactericidal.[4] Over the past few years we have been interested in the synthesis[5] and reactivity[6,7] of α-amino allenylphosphonates (AAP) **1**. In particular, in 2013, we have reported the chemo- and stereoselective reduction of α-AAPs **1** to α-amino vinylphosphonates **3** using H_2 and poisoned palladium catalyst (Scheme 1).[8] Our investigations in this field led us to postulate that the stereoselectivity of this reduction depends on the substitution pattern on the allenyl moiety whereas the chemoselectivity is due to the preferred reduction of the electron-richer, and in that case less hindered, double bond of the α-AAP **1**. Continuing our investigations in this field, we now report the chemoselective reduction of the same α-APPs **1** to prepare the corresponding α-amino allylphosphonates **2** using electrochemistry (Scheme 1).

Although the reduction of allenylphosphonates to vinylphosphonates is well documented using hydrogen and poisoned catalyst,[9,10] to date only one case of partial reduction of allenylphosphonates to allylphosphonates has been reported in the literature using ammonium formate and Pd catalyst.[11] In the case of α-amino allylphosphonates, after hydrolysis/transformation to the corresponding α-amino allylphosphonic acids, these compounds could be incorporated to peptides to induce antimicrobial properties.[12,13] To date the classical methods for the synthesis of α-amino allylphosphonates are based on a Kabachnik–Fields reaction followed by elimination.[14,15] A different strategy based on an aza-Wittig reaction was reported by Palacios and co-workers in 2007.[16] More recently, Zhang and co-workers have prepared allylphosphonates from a copper(I)-catalysed cyclopropene ring opening reaction.[17]



Scheme 1. Reduction of allenylphosphonates to vinyl- or allylphosphonates.

The common methods for the partial reduction of allenes involve the use of hydrogen with poisoned catalyst,[8–10] sodium in liquid ammonia to favour the *E* isomer[18] or in situ generated diimide,[19] the last ones having failed to reduce our substrates.[8] In this context, electrochemistry has been known for a long time to be efficient for the reduction of allenes.[20,21] The mechanism of such reduction has been studied as well.[22] More generally, there are several electrochemical methods well-known to oxidise and reduce organic compounds efficiently.[23] Electrochemistry has been defined by Frontana-Uribe and co-workers as a process that relates to 9 of the 12 principles of green chemistry by being clean, environmentally friendly and scalable to industrial level.[24] In the context of developing new selective transformations, we propose here to replace conventional reagents used in organic chemistry with electrodes in order to explore the electrolysis of α-amino allenylphosphonates **1** to obtain α-amino allylphosphonates **2**.

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Supporting information and ORCID(s) from the author(s) for this article are available on the WWW under https://doi.org/10.1002/ejoc.202000534.

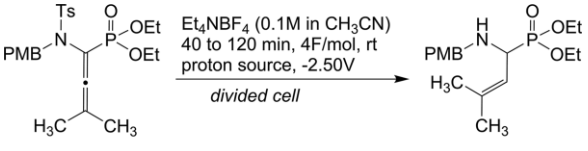
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Results and Discussion

For the optimisation of this electrochemical reduction, we started our investigation with the easily prepared α-AAP **4** (Table 1).[5] We selected acetonitrile as the solvent and Et_4NBF_4 as the supporting electrolyte due to their easiness of separation from the expected product. It has to be noted that other supporting electrolytes such as nBu_4NClO_4 and $LiClO_4$ were investigated as well; however, in that case the obtained results were less conclusive. First, we started to record the cyclic voltammogram for the α-AAP **4** (see Supporting Information). The reduction potential for the electrolysis was set to –2.50 V, slightly more negative than the peak potential of the reduction wave. Next, we turned our attention to the source of protons needed for this reduction. Although it has been known that the chosen electrolyte (Et_4NBF_4) could decompose during the electrolysis through a Hofmann type elimination to provide a proton and a molecule of ethylene,[21] we noticed in our case a longer reaction time along with the formation of multiple by-products leading to a more complicated separation of the expected α-amino allylphosphonate (α-AAllylP). Although the yield of formation of the α-AAllylP **5** is good in pure acetonitrile with no external source of protons (66 %), other products, under the same conditions, were obtained in a lower yield [see compounds **14** (55 %, Scheme 3), **16** (35 %, Scheme 4), **17** (37 %, Scheme 4), **28** (52 %, Scheme 6) and **33** (45 %, Scheme 6)]. Hence, we investigated different external proton sources (Table 1, entries 2–5) to conclude that in all cases the reaction is faster and cleaner than with the supporting electrolyte alone. The best results for the reduction of the α-AAP **4** to the α-AAllylP **5** were observed with phenol and diethyl malonate. The latter is electro-inactive and gives the best result in terms of reproducibility. Indeed, the use of succinimide (entry 2) provides the desired allylphosphonate **5** along with multiple by-products, whereas benzoic acid (entry 4), a stronger acid, leads to a low conversion of **4** to **5**. Moreover, the use of phenol (entry 3) is not recommended due to its toxicity and to the fact that traces of the latter are recovered in the final product.

Table 1. Optimisation of the reaction conditions: proton source.



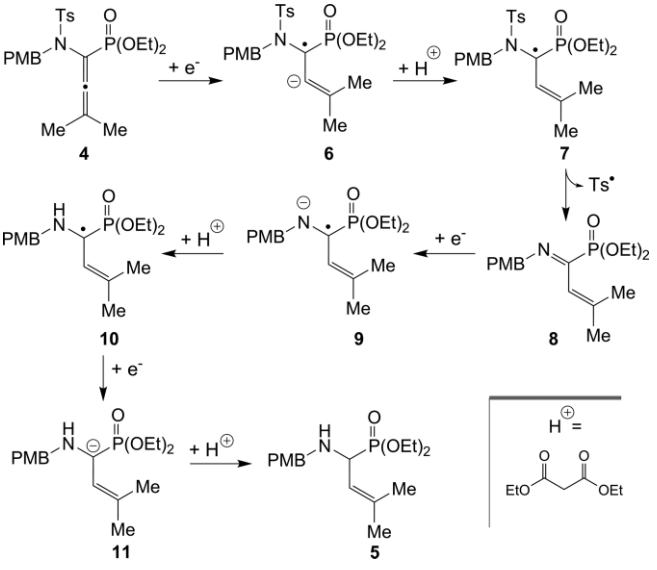
4 **5**

entry	proton source	conv.	yield
1	none	100%	66%
2	succinimide	75%	ND
3	phenol	100%	66%
4	$PhCO_2H$	9%	ND
5	diethyl malonate	100%	58% (68%)[a]

[a] Isolated yield obtained from the reduction of **4** in a 1:1 mixture of CH_3CN and DMF.

Although acetonitrile seems to be a good solvent for this reaction, we had some reproducibility problems as well as formation of multiple unidentified by-products, probably resulting from the decomposition of the starting material. By switching the solvent from acetonitrile to DMF we observed a more reproducible and cleaner reaction. In contrast with acetonitrile, in DMF, without any external source of protons, the conversion of the α-AAP **4** to the α-AAllylP **5** is incomplete and much longer than in acetonitrile. A good compromise was to use a mixture of CH_3CN and DMF (1:1). In these conditions, the α-AAP **4** was reduced to the α-AAllylP **5** with a good yield (68 %) without any traces of other organophosphorus by-products.

We propose the following mechanism for the formation of the α-AAllylP **5**. The addition of the first electron to α-AAP **4** provides the radical-anion **6**, which is protonated to form the radical **7**. Then a β-fragmentation takes place to cleave the *N*-Tosyl bond, generating the imine **8** and releasing the Ts radical.[25–28] The reduction of the imine **8** gives the new radical-anion **9** which is next protonated to give radical **10**. The latter is then reduced to the anion **11**, which is protonated to provide the desired α-AAllylP **5**. In this reduction a fourth election is needed to reduce the Ts radical generated earlier (Scheme 2).

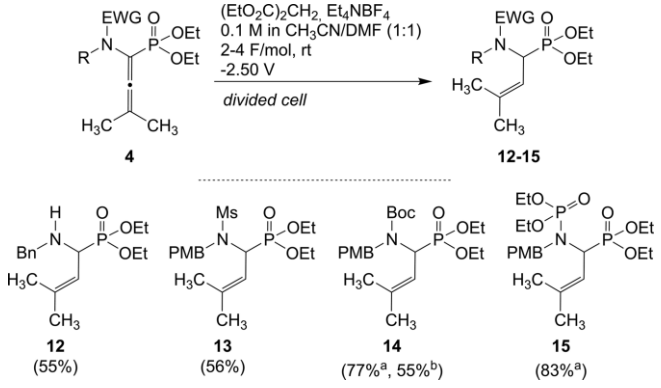


Scheme 2. Proposed mechanism for the reduction of the α-AAP **4** to the α-AAllylP **5**.

To confirm the formation of the radical **7**, leading to the cleavage of the N–Ts bond, we extended the scope of this reduction to other protecting groups on the nitrogen atom. Replacing the tosyl group with a *para*-methoxybenzene sulfonyl group led to the same type of cleavage and the resulting α-AAllylP **12** was formed in 55 % yield. However, in the same conditions the α-AAP protected with a mesyl group gave the protected α-AAllylP **13** in 56 % yield. In that case, only 2 F/mol and 2 equiv. of diethyl malonate were needed for this reduction. To force the cleavage of the *N*-Ms bond and to obtain the unprotected α-AAllylP **5**, instead of the protected α-AAllylP **13**, we applied a more negative potential for the reduction of the corresponding allene, successively at –2.50 V, –2.70 V and –2.90 V. In that case (–2.70 V and –2.90 V), we observed the formation of **5** as the minor product of this reaction and the formation of **13** as the major product. In both cases, the relative

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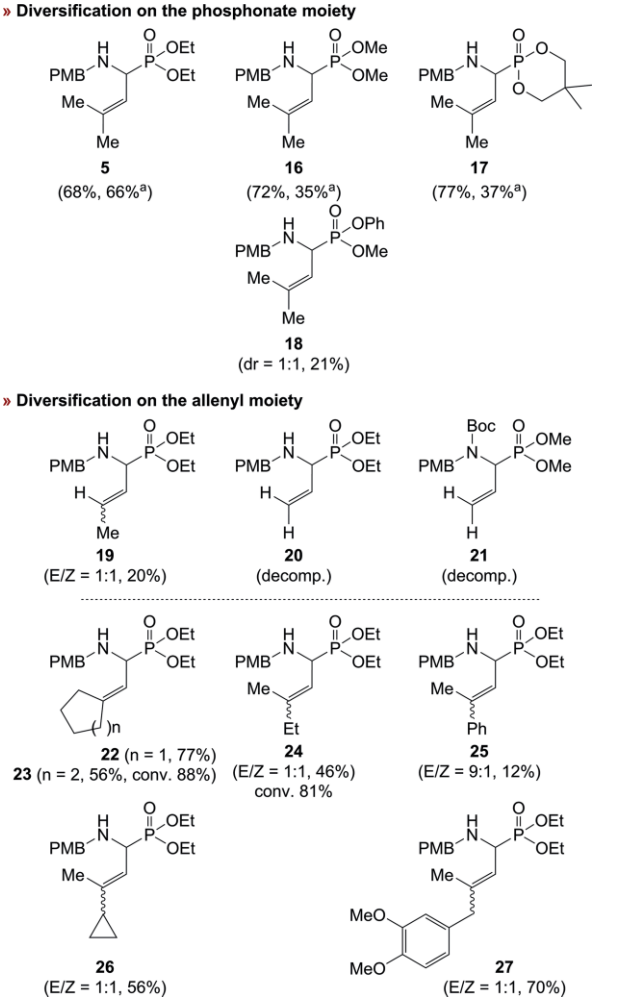
proportion of **5** vs. **13** is equal to 1:2. The reduction of the Boc-protected allenylphosphonate proceeds at a more negative potential (–2.70 V vs. SCE) and gives the corresponding α-AAllylP **14** in a high yield (77 %). Finally, the α-AAP protected as a phosphoramidate was reduced in the same conditions as the Boc-protected α-AAP (–2.70 V vs. SCE) to provide the α-AAllylP **15** in a high yield (83 %) (Scheme 3).



Scheme 3. Study on the effect of the protecting group on the nitrogen atom. [(a)] Reduction at –2.70 V vs. SCE. [(b)] Reduction in pure acetonitrile at –2.70 V vs. SCE without diethyl malonate.

Interested by the α-AAllylP **5**, and using our optimised conditions {Et_4NBF_4 [0.1 M in CH_3CN/DMF (1:1)], diethyl malonate (4.5 equiv.), 4 F/mol, –2.50 V, r.t.}, we next explored the scope and limitations for this reduction starting from the corresponding *N*-Tosyl α-AAP to give the resulting α-amino allylphosphonates **16**–**27** in moderate to excellent yields (Scheme 4).

By changing the nature of the alkyl group on the phosphonate moiety, we did not observe any significant shift of the reduction wave and therefore the applied potential for the conversion of the allenylphosphonates was kept at –2.50 V vs. SCE. The corresponding α-AAllyPs **5**, **16** and **17** were obtained in good yields (Scheme 4). The obtained yield for the formation of the α-AAllyP **5** in CH_3CN/DMF is higher than in pure CH_3CN (see Table 1). Changing one of the alkyl groups of the phosphonate moiety on the α-AAP **4** with a phenyl group gives after reduction (–2.50 V vs. SCE) the α-AAllylP **18**, in 21 % yield, as a 1:1 mixture of two inseparable diastereoisomers. The low yield observed for the formation of the α-AAllylP **18** is explained by the instability of the latter that decomposes during purification. Despite all of our efforts, the α-AAP **4**, with two phenyl groups on the phosphonate moiety, could not be prepared to explore its reduction to the corresponding α-AAllylP. The effect of the substitution on the allene moiety for this reduction was studied first by introducing only one methyl group in order to form the α-AAllylP **19**. In that case, we observed the formation of multiple products and the expected α-AAllylP **19** was isolated in a low yield (20 %) in a 1:1 mixture of isomers. This low yield is explained by the isomerisation of alkene to form the amino vinylphosphonate that later decomposed during the purification step. Next, we turned our attention to the reduction of non-substituted α-amino allenylphosphonates. The decomposition of the α-AAllylP **20** could be explained by the isomerisation of the carbon–carbon double bond to form the corresponding



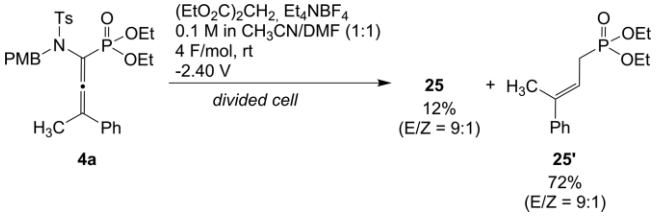
Scheme 4. Scope and limitations. Reaction conditions: $(EtO_2C)_2CH_2$ (4.5 equiv.), Et_4NBF_4 0.1 M in CH_3CN/DMF (1:1), r.t., 4 F/mol, –2.50V vs. SCE. Diastereoisomeric ratios (*dr*) are measured on the crude ^{31}P NMR. [(a)] The reduction was performed in pure CH_3CN without diethyl malonate.

α-amino vinylphosphonate. The latter is next decomposed leading to the formation of a pyruvate analogue.[29] In order to prevent this decomposition and based on our previous results with the Boc group, we tried to prepare the Boc-protected α-AAllylP **21**. However, the latter is decomposed under the reaction conditions. This result led us to continue to explore the scope of the reduction of the α-AAP **1** (R = Ts) bearing two substituents on the allene moiety. With identical substituents on the allene moiety, the corresponding α-AAllylPs **22**–**23** were formed in good yields. With different substituents on the allene moiety, we observed the formation of **24**, **26** and **27** in a 1:1 inseparable mixture of isomers. It is noteworthy that neither the size nor the electronic effects induced by the substituents could influence the stereoselectivity of the reduction (Scheme 4). The formation of the α-AAllylP **25** from the corresponding α-AAP **4a** was more complicated due to the presence of the aromatic ring on the allenyl moiety. In that case, the reduction potential was less negative (–2.40 V vs. SCE) due to the presence of multiple reduction waves on the starting material (see supporting information). However, despite that modification, the

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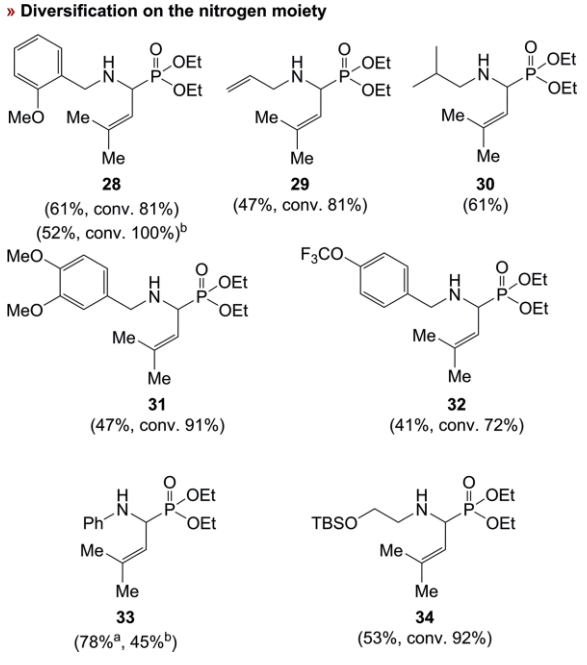
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α-AAllylP **25** was isolated in a low yield (12 %) (*E*/*Z* = 9:1 in favour of the *E* isomer) as an inseparable mixture of isomers. In this reduction, the major product is the allylphosphonate **25′** that is isolated in a high yield (72 %) as a 9:1 mixture of isomers in favour of the *E* isomer (Scheme 5). Changing the reduction potential to –2.30 V vs. SCE did not modify the ratio of the products formed in this reaction.



Scheme 5. Reduction of the α-AAP **4a** to the α-AAllylP **25** and the allylphosphonate **25′**.

Next, we turned our attention to the substitution on the nitrogen atom. Replacing the PMB group by an *ortho*-methoxybenzyl group led to the α-AAllylP **28** in a good yield (61 %) with an incomplete conversion of 81 %. On the other hand, alkenyl and alkyl substituents led respectively to the α-AAllylP **29** and **30** in a lower yield (47 % and 61 % respectively) partially explained by the incomplete conversion of the starting material (81 %) in the case of compound **29**. Increasing or decreasing the electron density on the aromatic ring was next explored to prepare the α-AAllylP **31** and **32** in moderate yields. We prepared the α-AAllylP **33** in a high yield (78 %) (Scheme 6). It has to be noted that in this case the reduction potential was set to –2.42 V vs. SCE due to the presence of a phenyl group on the nitrogen atom that anodically shifts the reduction wave of the corresponding α-AAP (see cyclic voltammogram in the Supporting Information). Finally, the compatibility of the silyl ether functionality, in these reaction conditions, was tested for the preparation of the α-AAllylP **34**. The latter was obtained in moderate yield (53 %) (Scheme 6).



Scheme 6. Scope and limitations: diversification on the nitrogen moiety. [a] Reduction at –2.42 V vs. SCE. [b] The reduction was performed in CH_3CN without diethyl malonate.

Conclusion

In conclusion, we have prepared a series of α-amino allylphosphonates through the reductive electrolysis of α-amino allenylphosphonates. This process is chemoselective for the formation of the allylphosphonates. Indeed, on the studied examples, no traces of vinyl or alkylphosphonates were observed after the reduction. In the case of α-amino allylphosphonates protected with a tosyl group, the latter is cleaved in the process. However, other protecting groups (mesylate, carbamate and phosphoramidate) have been successfully used to prepare the corresponding protected α-amino allylphosphonates. Moreover, with this new methodology in hand we now have access, from the same starting material (the α-amino allenylphosphonates) to either the α-amino vinylphosphonates as described in our previous work[8] or the α-amino allylphosphonates, as presented herein.

Experimental Section

All reactions were carried out under argon with magnetic stirring. All solvents and chemicals were purified based on standard procedures. Reagent grade solvents were used without purification for all extractions and work-up procedures. R_f values refer to values obtained by TLC on 0.25 mm silica gel plates (60 F254). Flash chromatography was carried out on silica gel 60 (35–70 μm) with various mixtures of ethyl acetate (EtOAc) and petroleum ether (PE), yields refer to chromatographically and spectroscopically pure compounds. All ^1H, ^{13}C and ^{31}P NMR spectra were recorded on 250, 300, 360, 400 MHz spectrometers at 300 K. Chemical shifts were given in parts per million (ppm, δ), referenced to the solvent peak of $CDCl_3$, defined at δ = 7.26 ppm (^1H NMR) and δ = 77.16 (^{13}C NMR). HRMS spectra were recorded with a MicroTOFq spectrometer.

Electrochemical data were obtained using an EG & G 273A potentiostat driven by a PC with the M270 software. A divided cell with a standard three-electrode configuration was used for cyclic voltammetry experiments. The reference electrode was a saturated calomel electrode (SCE) and the counter electrode a platinum gauze of large surface area; both electrodes were separated from the bulk electrolyte solution via fritted compartments filled with the same electrolyte, which was a solution of Et_4NBF_4 (0.1 M) in a 1:1 mixture of anhydrous DMF and anhydrous CH_3CN. The working electrode was a ca. $120 \times 10 \times 2$ mm^3 stick (GC, MERSEN, France) both for cyclic voltammetry and for controlled potential coulometry experiments. Prior to each experiment, solutions were de-aerated thoroughly for at least 15 min with pure Ar. A positive pressure of this gas was maintained during subsequent work. All cyclic voltammograms were recorded at a scan rate of 100 mV s^{-1} unless otherwise stated. Before the electrolysis, a cyclic voltammogram of the allene to be reduced, mixed with the diethyl malonate, was recorded in the same conditions as the electrolysis (same electrodes, vessels, solvents, temperature) and is represented in the following CVs as the blue curve. After the electrolysis, a new cyclic voltammogram (red curve) of the reaction mixture was recorded. All experiments

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were performed at room temperature, which was always close to 25 °C.

General Procedure for the Electrochemical Reduction of the α-AAP

In a divided cell, the *N*-Tosyl α-AAP (0.215 mmol) and diethyl malonate (0.966 mmol, 4.5 equiv.) were added to 25 mL of a solution of Et_4NBF_4 (0.1 M) in a 1:1 mixture of anhydrous DMF and anhydrous CH_3CN. The solution was degassed under argon for 15 min before electrolysis. A constant potential of –2.50 V vs. SCE (unless otherwise mentioned) was then applied at room temperature. The argon degassing was maintained during the electrolysis. When the current was less than 5 % of its starting value, usually after 4–5 F/mol of starting material (5 h), the electrolysis was stopped. The solvent was removed under vacuo and the residue was washed with EtOAc. The resulting suspension was filtered through a pad of Celite and the solvent was evaporated under vacuo. The residue was purified by silica gel chromatography to yield the expected α-AAllylP.

Diethyl 1-(4-Methoxybenzylamino)-3-methylbut-2-enylphosphonate (5): Prepared according to the general procedure **A**. Flash Chromatography: EtOAc/PE = 50:50 to 100:0; yellow oil, 54 mg, 68 % yield; R_f = 0.25 (EtOAc = 100); ^{1}H NMR (300 MHz, $CDCl_3$) δ = 7.16 (d, *J* = 8.5 Hz, 2H), 6.80 (d, *J* = 8.6 Hz, 2H), 5.14–4.99 (m, 1H), 4.22–3.99 (m, 4H), 3.80 (d, *J* = 13.0 Hz, 1H), 3.75 (s, 3H), 3.70–3.49 (m, 2H), 1.85 (br. s, 1H), 1.77 (d, *J* = 4.8 Hz, 3H), 1.54 (d, *J* = 3.6 Hz, 3H), 1.27 (t, *J* = 7.0 Hz, 3H), 1.25 (t, *J* = 7.0 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ = 158.6, 138.4 (d, *J* = 11.3 Hz), 131.7, 129.4, 119.8 (d, *J* = 5.7 Hz), 113.7, 62.5, 55.2, 53.0 (d, *J* = 158.3 Hz), 50.5 (d, *J* = 17.1 Hz), 26.0, 18.6, 16.5 (d, *J* = 3.6 Hz); ^{31}P NMR (121 MHz, $CDCl_3$) δ = 25.4; HRMS (ESI) *m/z* [MNa^+] calcd. for $C_{17}H_{28}NNaO_4P$: 364.1648, found 364.1632.

Diethyl [1-(Benzylamino)-3-methylbut-2-en-1-yl]phosphonate (12): Prepared according to the general procedure A. Flash Chromatography: EtOAc 100; yellow oil, 44 mg, 55 % yield; R_f = 0.35 (EtOAc = 100); ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.37–7.18 (m, 5H), 5.15–5.05 (m, 1H), 4.24–4.04 (m, 4H), 3.91 (d, *J* = 13.4 Hz, 1H), 3.75–3.62 (m, 2H), 1.86 (br. s, 1H), 1.81 (dd, *J* = 4.9, 1.2 Hz, 3H), 1.56 (dd, *J* = 3.6, 1.2 Hz, 3H), 1.32 (d, *J* = 7.2 Hz, 3H), 1.30 (d, *J* = 7.2 Hz, 3H). ^{13}C NMR (91 MHz, $CDCl_3$) δ = 139.7, 138.6 (d, *J* = 13.5 Hz), 128.3, 127.0, 119.7 (d, *J* = 6.1 Hz), 62.5, 53.1 (d, *J* = 158.5 Hz), 51.1 (d, *J* = 17.1 Hz), 26.1, 18.6, 16.5; ^{31}P NMR (121 MHz, $CDCl_3$) δ = 25.4; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{17}H_{27}NO_3P$: 324.1723, found 324.1720.

Diethyl 1-[*N*-(4-Methoxybenzyl)methylsulfonamido]-3-methylbut-2-enylphosphonate (13): Prepared according to the general procedure A. Flash Chromatography: EtOAc/PE = 80:20; colorless oil, 6 mg, 55 % yield; R_f = 0.53 (EtOAc = 100); ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.45 (d, *J* = 8.6 Hz, 2H), 6.87 (d, *J* = 8.6 Hz, 2H), 5.42–5.30 (m, 1H), 5.12 (dd, *J* = 22.6, 10.2 Hz, 1H), 4.65 (d, *J* = 15.2 Hz, 1H), 4.50 (d, *J* = 15.2 Hz, 1H), 4.23–4.07 (m, 4H), 3.82 (s, 3H), 2.68 (s, 3H), 1.80–1.71 (m, 6H), 1.35 (t, *J* = 7.1 Hz, 3H), 1.31 (d, *J* = 7.1 Hz, 3H); ^{13}C NMR (101 MHz, $CDCl_3$) δ = 159.3, 141.4 (d, *J* = 14.1 Hz), 130.7, 129.3, 115.6, 113.8, 63.1 (d, *J* = 7.0 Hz), 62.7 (d, *J* = 7.2 Hz), 55.4, 53.2 (d, *J* = 160.7 Hz), 49.3, 41.6, 26.2, 18.5, 16.6; ^{31}P NMR (101 MHz, $CDCl_3$) δ = 22.3; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{18}H_{31}NO_6PS$: 420.1604, found 420.1592.

***tert*-Butyl 1-(Diethoxyphosphoryl)-3-methylbut-2-enyl(4-methoxybenzyl)carbamate (14):** Prepared according to the general procedure A. Flash Chromatography: EtOAc/PE = 40:60 to 80:20; colorless oil, 77 mg, 77 % yield; R_f = 0.68 (EtOAc = 100); ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.13 (br. d, *J* = 7.2 Hz, 2H), 6.83 (d, *J* = 8.6 Hz, 2H), 5.60–5.30 (m, 2H), 4.52 (dd, *J* = 45.6, 16.0 Hz, 2H), 4.14 (dd, *J* = 14.4, 7.2 Hz, 4H), 3.81 (s, 3H), 1.73 (br. d, *J* = 23.4 Hz, 6H), 1.58–1.46 (m, 3H), 1.39–1.24 (m, 12H); ^{13}C NMR (63 MHz, $CDCl_3$) δ = 158.2, 155.4, 141.0 (d, *J* = 15.2 Hz), 132.3, 127.8, 115.8, 113.3, 80.3, 62.5 (d, *J* = 7.3 Hz), 62.3 (d, *J* = 7.3 Hz), 55.3, 50.3 (d, *J* = 150.2 Hz), 48.0, 28.2, 25.8, 18.8, 16.5; ^{31}P NMR (101 MHz, $CDCl_3$) δ = 23.9; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{22}H_{37}NO_6P$: 442.2353, found 442.2346.

Diethyl [1-(Diethoxyphosphoryl)-3-methylbut-2-en-1-yl]-(4-methoxybenzyl)phosphoramidate (15): Prepared according to the general procedure A. Flash Chromatography: E/MeOH = 100:0 to 97:3; colorless oil, 118 mg, 83 % yield; R_f = 0.27 (DCM/MeOH = 95:5); ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.16 (d, *J* = 8.7 Hz, 2H), 6.61 (d, *J* = 8.8 Hz, 2H), 5.19–5.09 (m, 1H), 4.52 (ddd, *J* = 22.8, 12.5, 10.5 Hz, 1H), 4.25–4.04 (m, 2H), 3.96–3.63 (m, 8H), 3.57 (s, 3H), 1.47 (dd, *J* = 3.4, 1.2 Hz, 3H), 1.44 (dd, *J* = 2.5, 1.4 Hz, 3H), 1.14–0.95 (m, 12H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.7, 138.75 (d, *J* = 14.3 Hz), 131.4, 130.2, 117.5, 113.2, 62.4 (d, *J* = 7.0 Hz), 62.0 (d, *J* = 4.2 Hz), 55.2, 52.5 (dd, *J* = 158.8, 5.3 Hz), 48.5 (d, *J* = 2.9 Hz), 25.9, 18.2, 16.6–16.2 (m), 16.2–15.8 (m); ^{31}P NMR (101 MHz, $CDCl_3$) δ = 24.2 (d, *J* = 22.1 Hz), 9.5 (d, *J* = 22.2 Hz); HRMS (ESI) *m/z* [MNa^+] calcd. for $C_{21}H_{37}NNaO_7P_2$: 500.1937, found 500.1918.

Dimethyl 1-(4-Methoxybenzylamino)-3-methylbut-2-enylphosphonate (16): Prepared according to the general procedure A. Flash Chromatography: EtOAc/PE = 80:20 to 90:10; yellow oil, 58 mg, 72 % yield; R_f = 0.34 (DCM/MeOH = 96:4); ^{1}H NMR (300 MHz, $CDCl_3$) δ = 7.20 (d, *J* = 8.6 Hz, 2H), 6.85 (d, *J* = 8.6 Hz, 2H), 5.15–5.05 (m, 1H), 3.90–3.56 (m, 12H), 1.87 (br. s, 1H), 1.82 (d, *J* = 4.8 Hz, 3H), 1.58 (dd, *J* = 3.5, 1.1 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ = 158.7, 138.9 (d, *J* = 13.6 Hz), 131.6, 129.5, 119.4 (d, *J* = 6.5 Hz), 113.7, 55.3, 53.4 (d, *J* = 7.1 Hz), 53.3 (d, *J* = 7.0 Hz), 52.7 (d, *J* = 157.5 Hz), 50.6 (d, *J* = 17.0 Hz), 26.1 (d, *J* = 2.2 Hz), 18.6 (d, *J* = 1.5 Hz); ^{31}P NMR (121 MHz, $CDCl_3$) δ = 27.6; HRMS (ESI) *m/z* [MNa^+] calcd. for $C_{15}H_{24}NNaO_4P$: 336.1335, found 336.1318.

1-(5,5-Dimethyl-2-oxido-1,3,2-dioxaphosphinan-2-yl)-*N*-(4-methoxybenzyl)-3-methylbut-2-en-1-amine (17): Prepared according to the general procedure A. Flash Chromatography: EtOAc/PE = 80:20 to 100:0; colorless oil, 55 mg, 77 % yield; R_f = 0.13 (EtOAc = 100); ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.29 (t, *J* = 51.2 Hz, 2H), 6.84 (d, *J* = 8.6 Hz, 2H), 5.20–5.09 (m, 1H), 4.24–4.10 (m, 2H), 3.91–3.72 (m, 7H), 3.65 (d, *J* = 13.2 Hz, 1H), 2.04 (br. s, 1H), 1.80 (dd, *J* = 5.0, 1.0 Hz, 3H), 1.58 (dd, *J* = 3.7, 1.0 Hz, 3H), 1.07 (s, 3H), 0.99 (s, 3H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.7, 139.1 (d, *J* = 13.6 Hz), 131.4, 129.5, 119.2 (d, *J* = 6.9 Hz), 113.7, 75.4 (t, *J* = 7.2 Hz), 55.2, 52.4 (d, *J* = 155.1 Hz), 50.6 (d, *J* = 16.0 Hz), 32.7, 26.0, 21.5, 18.7; ^{31}P NMR (101 MHz, $CDCl_3$) δ = 21.4; HRMS (ESI) *m/z* [MNa^+] calcd. for $C_{18}H_{28}NNaO_4P$: 376.1648, found 376.1631.

Ethyl Phenyl {1-[(4-Methoxybenzyl)amino]-3-methylbut-2-en-1-yl}phosphonate (18): Prepared according to the general procedure A. Flash Chromatography: EtOAc/PE = 40:60; yellow oil, 24 mg, 21 % yield; R_f = 0.59 (EtOAc = 100); The following description takes into consideration the mixture of the two stereoisomers enriched in one of them ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.35–7.29 (m, 2H), 7.25–7.12 (m, 5H), 6.88–6.82 (m, 2H), 5.21–5.14 (m, 1H), 4.25–4.14 (m, 2H), 3.92–3.84 (m, 2H), 3.83–3.80 (m, 3H), 3.67 (d, *J* = 13.2 Hz, 1H), 1.83 (dd, *J* = 5.0, 1.3 Hz, 3H), 1.72 (br s, 1H), 1.57 (dd, *J* = 3.8, 1.3 Hz, 3H), 1.31–1.25 (m, 3H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.7, 139.1 (d, *J* = 14.2 Hz), 131.6, 130.1, 129.5, 128.4, 124.6, 120.5 (d, *J* = 4.1 Hz), 119.4 (d, *J* = 6.5 Hz), 113.7, 63.4 (d, *J* = 7.2 Hz), 55.3, 53.1 (d, *J* = 159.0 Hz), 50.5, 26.0, 18.7, 16.4; ^{31}P NMR (101 MHz, $CDCl_3$) δ = 22.4, 22.1; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{21}H_{29}NO_4P$: 390.1829, found 390.1810.

(*E*,*Z*)-Diethyl {1-[(4-Methoxybenzyl)amino]but-2-en-1-yl}phosphonate (19): Prepared according to the general procedure A. Flash

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Chromatography: CH_2Cl_2/MeOH = 100:0 to 99:1; colorless oil, 20 mg, 20 % yield; R_f = 0.31 (EtOAc = 100); The following description takes into consideration the 1:1 mixture of the two stereoisomers ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.25–7.19 (m, 4H), 6.89–6.83 (m, 4H), 5.91–5.66 (m, 2H), 5.46–5.30 (m, 2H), 4.22–4.07 (m, 8H), 3.91–3.77 (m, 2H + 1H), 3.81 (s, 6H), 3.63 (dd, *J* = 13.2, 7.1 Hz, 2H), 3.44 (dd, *J* = 18.2, 8.6 Hz, 1H), 1.82–1.76 (m, 3H + 2H from NH bond), 1.60 (ddd, *J* = 6.9, 4.2, 1.8 Hz, 3H), 1.36–1.28 (m, 12H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.7, 131.7, 131.6, 131.2, 131.0, 130.3, 130.1, 129.5, 129.5, 125.7 (d, *J* = 6.8 Hz), 125.5 (d, *J* = 6.0 Hz), 113.7, 62.6 (s, *J* = 8.1 Hz), 62.5 (d, *J* = 8.1 Hz), 57.3 (d, *J* = 155.9 Hz), 55.3, 51.8 (d, *J* = 157.6 Hz), 50.6 (d, *J* = 17.4 Hz), 50.5 (d, *J* = 15.4 Hz), 18.0 (br. s), 16.6–16.3 (m), 13.6 (br. s); ^{31}P NMR (101 MHz, $CDCl_3$) δ = 24.9, 24.8; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{16}H_{27}NO_4P$: 328.1672, found 328.1665.

Diethyl {2-Cyclopentylidene-1-[(4-methoxybenzyl)amino]ethyl}-phosphonate (22): Prepared according to the general procedure A. Flash Chromatography: EtOAc 100; yellow oil, 65 mg, 77 % yield; R_f = 0.17 (EtOAc/PE = 90:10); ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.12 (d, *J* = 8.7 Hz, 2H), 6.77 (d, *J* = 8.7 Hz, 2H), 5.12 (dtd, *J* = 7.0, 4.6, 2.2 Hz, 1H), 4.13–3.98 (m, 4H), 3.78 (d, *J* = 13.2 Hz, 1H), 3.72 (s, 3H), 3.55 (d, *J* = 13.5 Hz, 1H), 3.53–3.43 (m, 1H), 2.32–2.21 (m, 2H), 2.14–2.02 (m, 1H), 1.99–1.86 (m, 1H), 1.80 (br. s, 1H), 1.63–1.50 (m, 4H), 1.24 (t, *J* = 7.2 Hz, 3H), 1.22 (t, *J* = 7.2 Hz, 3H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.6, 150.4 (d, *J* = 13.5 Hz), 131.8, 129.4, 115.0 (d, *J* = 7.2 Hz), 113.7, 62.5 (d, *J* = 6.8 Hz), 55.2, 54.7 (d, *J* = 158.0 Hz), 50.5 (d, *J* = 17.1 Hz), 33.9, 29.3, 26.2 (d, *J* = 8.1 Hz), 16.5 (d, *J* = 4.7 Hz); ^{31}P NMR (121 MHz, $CDCl_3$) δ = 25.6; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{19}H_{31}NO_4P$: 368.1985, found 368.1968.

Diethyl {2-Cyclohexylidene-1-[(4-methoxybenzyl)amino]ethyl}-phosphonate (23): Prepared according to the general procedure A. Flash Chromatography: EtOAc 100; yellow oil, 48 mg, 56 % yield; R_f = 0.19 (EtOAc = 100); ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.13 (d, *J* = 8.6 Hz, 2H), 6.77 (d, *J* = 8.6 Hz, 2H), 4.96 (dd, *J* = 10.0, 5.2 Hz, 1H), 4.11–4.00 (m, 4H), 3.83–3.60 (m, 5H), 3.54 (d, *J* = 13.2 Hz, 1H), 2.10 (br. s, 2H), 1.96 (br. s, 2H), 1.84 (br. s, 1H), 1.54–1.39 (m, 6H), 1.24 (d, *J* = 7.1 Hz, 3H), 1.23–1.20 (m, 3H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.7, 146.3 (d, *J* = 13.5 Hz), 131.7, 129.5, 116.3 (d, *J* = 5.8 Hz), 113.7, 62.6 (d, *J* = 6.9 Hz), 62.4 (d, *J* = 6.9 Hz), 55.3, 51.9 (d, *J* = 158.9 Hz), 50.4 (d, *J* = 17.1 Hz), 37.3, 29.6, 28.6, 27.6, 26.6, 16.5 (d, *J* = 6.0 Hz); ^{31}P NMR (121 MHz, $CDCl_3$) δ = 25.2; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{20}H_{33}NO_4P$: 382.2142, found 382.2126.

Diethyl {1-[(4-Methoxybenzyl)amino]-3-methylpent-2-en-1-yl}-phosphonate (24): Prepared according to the general procedure A. Flash Chromatography: EtOAc 100; colorless oil, 43 mg, 46 % yield; conv. 81 %; R_f = 0.29 [EtOAc (5 vol.-% NH_3) = 100]; ^{1}H NMR (300 MHz, $CDCl_3$) δ = 7.25–7.16 (m, 2 × 2H), 6.89–6.81 (m, 2 × 2H), 5.14–5.01 (m, 2 × 1H), 4.24–4.03 (m, 2 × 4H), 3.89–3.57 (m, 2 × 3H + 2 × 2H + 2 × 1H), 2.16–2.04 (m, 2H), 2.03–1.92 (m, 2H), 1.80 (dd, *J* = 4.7, 1.3 Hz, 3H), 1.57 (dd, *J* = 3.7, 1.3 Hz, 3H), 1.39–1.24 (m, 2 × 6H), 1.05 (t, *J* = 7.4 Hz, 3H), 0.96 (t, *J* = 7.6 Hz, 3H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.6, 143.9 (d, *J* = 12.2 Hz), 131.8 (d, *J* = 8.1 Hz), 129.5, 129.4, 119.3 (d, *J* = 6.1 Hz), 118.3 (d, *J* = 6.4 Hz), 113.7, 62.8–62.2 (m), 55.3, 52.9 (d, *J* = 158.3 Hz), 52.8 (d, *J* = 158.3 Hz), 50.5 (d, *J* = 16.6 Hz), 50.4 (d, *J* = 16.4 Hz), 32.6, 25.4, 23.0 (d, *J* = 1.8 Hz), 16.9, 16.5 (d, *J* = 2.3 Hz), 16.5 (d, *J* = 2.0 Hz), 12.6 (d, *J* = 3.1 Hz), 12.4 (d, *J* = 2.6 Hz); ^{31}P NMR (101 MHz, $CDCl_3$) δ = 25.4, 25.2; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{18}H_{31}NO_4P$: 356.1985, found 356.1973.

Diethyl {1-[(4-Methoxybenzyl)amino]-3-phenylbut-2-en-1-yl}-phosphonate (25): Prepared according to the general procedure A. Flash Chromatography: EtOAc (5 vol.-% NH_3)/PE 80:20; yellow oil, 12 mg, 12 % yield; R_f = 0.25 [EtOAc (5 vol.-% NH_3) = 100]; Major isomer (*E*) ^{1}H NMR (300 MHz, $CDCl_3$) δ = 7.48–7.19 (m, 7H), 7.02–6.68 (m, 2H), 5.77–5.66 (m, 1H), 4.27–4.02 (m, 4H), 3.97–3.65 (m, 3H), 3.82 (s, 3H), 2.00 (dd, *J* = 4.0, 1.4 Hz, 3H), 1.87 (br. s, 1H), 1.38–1.26 (m, 6H); Major isomer (*E*) ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.9, 143.0, 140.7 (d, *J* = 13.4 Hz), 131.6, 129.7, 128.5, 127.6, 126.0, 123.4 (d, *J* = 7.1 Hz), 113.9, 62.9 (d, *J* = 6.9 Hz), 55.4, 53.8 (d, *J* = 158.1 Hz), 50.9 (d, *J* = 17.6 Hz), 16.9, 16.7 (d, *J* = 5.7 Hz); ^{31}P NMR (101 MHz, $CDCl_3$) δ = 25.3 (min.), 24.6 (maj.); HRMS (ESI) *m/z* [MH^+] calcd. for $C_{22}H_{31}NO_4P$: 404.1985, found 404.1975.

Diethyl (3-Phenylbut-2-en-1-yl)phosphonate (25′): Flash Chromatography: EtOAc (5 vol.-% NH_3)/PE 80:20; yellow oil, 48 mg, 72 % yield; R_f = 0.34 [EtOAc (5 vol.-% NH_3) = 100]; Major isomer (*E*) ^{1}H NMR (300 MHz, $CDCl_3$) δ = 7.43–7.18 (m, 1H), 5.84–5.72 (m, 1H), 4.20–4.03 (m, 1H), 2.78 (dd, *J* = 22.7, 7.9 Hz, 1H), 2.14–2.03 (m, 1H), 1.32 (t, *J* = 7.1 Hz, 1H); Major isomer (*E*) ^{13}C NMR (75 MHz, $CDCl_3$) δ = 143.1 (d, *J* = 3.4 Hz), 139.3 (d, *J* = 14.7 Hz), 128.2, 127.1, 125.8, 116.2 (d, *J* = 11.5 Hz), 61.9 (d, *J* = 6.7 Hz), 27.4 (d, *J* = 140.1 Hz), 16.5 (d, *J* = 5.9 Hz), 16.2; ^{31}P NMR (121.5 MHz, $CDCl_3$) δ = 28.3 (min.), 27.6 (maj.); HRMS (ESI) *m/z* [MH^+] calcd. for $C_{14}H_{22}O_3P$: 269.1301, found 269.1290.

Diethyl {3-Cyclopropyl-1-[(4-methoxybenzyl)amino]but-2-en-1-yl}phosphonate (26): Prepared according to the general procedure A. Flash Chromatography: EtOAc 100; yellow oil, 49 mg, 56 % yield; R_f = 0.11 (EtOAc = 100); Analytical data for the mixture of both isomers: ^{1}H NMR (300 MHz, $CDCl_3$) δ = 7.32–7.14 (m, 4H), 6.84 (d, *J* = 8.4 Hz, 4H), 5.24–5.07 (m, 2H), 4.24–4.04 (m, 8H), 4.01–3.53 (m, 13H), 1.85 (s, 2H), 1.50 (dd, *J* = 4.8, 1.1 Hz, 3H), 1.48–1.46 (m, 1H), 1.46–1.42 (m, 3H), 1.39–1.35 (m, 1H), 1.35–1.25 (m, 12H), 0.66–0.54 (m, 6H), 0.54–0.45 (m, 2H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.6, 142.6 (d, *J* = 13.4 Hz), 141.7 (d, *J* = 13.3 Hz), 131.8, 131.7, 129.6, 129.5, 120.6 (d, *J* = 6.0 Hz), 117.9 (d, *J* = 6.3 Hz), 113.7, 113.6, 62.5 (d, *J* = 11.8 Hz), 55.2, 52.8 (d, *J* = 158.5 Hz), 52.1 (d, *J* = 158.5 Hz), 50.5, 50.3, 19.0, 16.5 (d, *J* = 5.5 Hz), 14.8, 12.9, 5.0, 4.7, 4.5; ^{31}P NMR (101 MHz, $CDCl_3$) δ = 25.4, 25.3; HRMS (ESI) *m/z* [MNa^+] calcd. for $C_{19}H_{30}NO_4P$: 390.1805, found 390.1791.

Diethyl {4-(3,4-Dimethoxyphenyl)-1-[(4-methoxybenzyl)amino]-3-methylbut-2-en-1-yl}phosphonate (27): Prepared according to the general procedure A. Flash Chromatography: EtOAc 100; yellow oil, 50 mg, 70 % yield; R_f = 0.19 (isomer *Z*), 0.11 (isomer *E*) (EtOAc = 100); Analytical data for the *Z* isomer: ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.15 (d, *J* = 8.7 Hz, 2H), 6.77 (d, *J* = 8.7 Hz, 2H), 6.73–6.58 (m, 3H), 5.25–5.16 (m, 1H), 4.16–4.00 (m, 4H), 3.88–3.81 (m, 2H), 3.78 (s, 3H), 3.78 (s, 3H), 3.72 (s, 3H), 3.59 (d, *J* = 13.0 Hz, 1H), 3.28–3.12 (m, 2H), 1.73 (br. s, 1H), 1.64 (dd, *J* = 4.6, 1.3 Hz, 3H), 1.26 (t, *J* = 7.0 Hz, 3H), 1.25 (t, *J* = 7.0 Hz, 3H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.7, 148.8, 147.4, 141.0 (d, *J* = 13.5 Hz), 131.7 (d, *J* = 5.3 Hz), 129.3, 121.4 (d, *J* = 5.7 Hz), 120.7, 113.8, 112.2, 111.1, 62.7 (d, *J* = 7.1 Hz), 62.5 (d, *J* = 6.6 Hz), 55.9, 55.3, 53.3 (d, *J* = 158.2 Hz), 50.8 (d, *J* = 16.0 Hz), 38.0, 23.6, 16.6 (d, *J* = 5.6 Hz); Analytical data for the *E* isomer: ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.10 (d, *J* = 8.7 Hz, 2H), 6.79–6.58 (m, 5H), 5.21–5.13 (m, 1H), 4.15–3.96 (m, 4H), 3.87–3.81 (m, 1H), 3.79 (s, 3H), 3.78 (s, 3H), 3.72 (s, 3H), 3.69–3.52 (m, 2H), 3.26 (d, *J* = 3.4 Hz, 2H), 1.76 (br. s, 1H), 1.43 (dd, *J* = 3.7, 1.3 Hz, 3H), 1.22 (t, *J* = 7.2 Hz, 3H), 1.20 (t, *J* = 7.2 Hz, 3H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 158.7, 148.9, 147.5, 141.6 (d, *J* = 13.4 Hz), 132.1, 131.5, 129.5, 121.6 (d, *J* = 6.6 Hz), 120.9, 113.7, 111.9, 111.1, 62.6 (d, *J* = 7.0 Hz), 62.5 (d, *J* = 6.8 Hz), 55.9, 55.8, 55.3, 52.9 (d, *J* = 158.0 Hz), 50.6 (d, *J* = 17.4 Hz), 46.0, 16.7, 16.5 (d, *J* = 5.4 Hz); ^{31}P NMR (101 MHz, $CDCl_3$) δ = 25.1, 25.0; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{25}H_{37}NO_6P$: 478.2353, found 478.2341.

Diethyl 1-(2-Methoxybenzylamino)-3-methylbut-2-enylphosphonate (28): Prepared according to the general procedure A. Flash Chromatography: EtOAc 100; yellow oil, 49 mg, 61 % yield; conv.

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81 %; R_f = 0.23 (EtOAc = 100); ^{1}H NMR (300 MHz, $CDCl_3$) δ = 7.22 (td, *J* = 7.9, 1.7 Hz, 1H), 7.13 (dd, *J* = 7.3, 1.6 Hz, 1H), 6.92–6.81 (m, 2H), 5.14–5.06 (m, 1H), 4.19–4.04 (m, 4H), 3.95 (d, *J* = 13.9 Hz, 1H), 3.82 (s, 3H), 3.68 (d, *J* = 10.2 Hz, 1H), 3.63 (d, *J* = 8.3 Hz, 1H), 2.23 (br s, 1H), 1.79 (dd, *J* = 5.1, 1.0 Hz, 3H), 1.54 (dd, *J* = 3.9, 1.2 Hz, 3H), 1.29 (t, *J* = 7.1 Hz, 3H), 1.28 (t, *J* = 7.1 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ = 157.9, 138.4 (d, *J* = 13.1 Hz), 130.2, 128.3, 127.3, 120.2, 119.6 (d, *J* = 8.2 Hz), 110.2, 62.4 (d, *J* = 6.8 Hz), 55.1, 53.1 (d, *J* = 158.1 Hz), 47.0 (d, *J* = 17.4 Hz), 26.1, 18.6, 16.5 (d, *J* = 5.3 Hz); ^{31}P NMR (121 MHz, $CDCl_3$) δ = 25.3; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{17}H_{29}NO_4P$: 342.1829, found 342.1815.

Diethyl [1-(Allylamino)-3-methylbut-2-en-1-yl]phosphonate (29): Prepared according to the general procedure A. Flash Chromatography: EtOAc 100; yellow oil, 36 mg, 47 % yield; conv. 81 %; R_f = 0.25 (CH_2Cl_2/MeOH = 95:5); ^{1}H NMR (360 MHz, $CDCl_3$) δ = 5.84 (dddd, *J* = 17.1, 10.3, 6.8, 5.3 Hz, 1H), 5.20–5.01 (m, 3H), 4.22–4.08 (m, 4H), 3.81–3.68 (m, 1H), 3.35 (dd, *J* = 14.1, 5.1 Hz, 1H), 3.13 (dd, *J* = 14.1, 6.8 Hz, 1H), 1.80 (dd, *J* = 4.7, 1.1 Hz, 3H), 1.68 (dd, *J* = 3.6, 1.2 Hz, 3H), 1.70 (br. s, 1H), 1.33 (d, *J* = 7.2 Hz, 3H), 1.31 (d, *J* = 7.2 Hz, 3H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 138.5 (d, *J* = 13.9 Hz), 136.6, 119.9 (d, *J* = 6.0 Hz), 116.5, 62.6 (d, *J* = 6.8 Hz), 53.5 (d, *J* = 158.4 Hz), 50.0 (d, *J* = 16.7 Hz), 26.1, 18.7, 16.6 (d, *J* = 5.2 Hz); ^{31}P NMR (121 MHz, $CDCl_3$) δ = 25.5; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{12}H_{25}NO_3P$: 262.1567, found 262.1557.

Diethyl [1-(Isobutylamino)-3-methylbut-2-en-1-yl]phosphonate (30): Prepared according to the general procedure A. Flash Chromatography: EtOAc 100; yellow oil, 65 mg, 61 % yield; R_f = 0.29 (CH_2Cl_2/MeOH = 95:5); ^{1}H NMR (300 MHz, $CDCl_3$) δ = 5.12–5.01 (m, 1H), 4.24–4.05 (m, 4H), 3.70 (dd, *J* = 17.3, 9.9 Hz, 1H), 2.47 (dd, *J* = 11.4, 7.1 Hz, 1H), 2.32 (dd, *J* = 11.5, 6.4 Hz, 1H), 1.89 (s, 1H), 1.79 (dd, *J* = 4.7, 1.2 Hz, 3H), 1.74–1.63 (m, 4H), 1.38–1.22 (m, 6H), 0.90 (d, *J* = 0.9 Hz, 3H), 0.88 (d, *J* = 1.0 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ = 138.0 (d, *J* = 14.0 Hz), 120.0 (d, *J* = 5.9 Hz), 62.6 (d, *J* = 7.3 Hz), 62.5 (d, *J* = 7.2 Hz), 55.7, 54.8 (d, *J* = 171.2 Hz), 28.2, 26.0, 20.7, 20.5, 18.6, 16.5 (br. s); ^{31}P NMR (121 MHz, $CDCl_3$) δ = 25.0; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{13}H_{29}NO_3P$: 278.1880, found 278.1879.

Diethyl {1-[(3,4-Dimethoxybenzyl)amino]-3-methylbut-2-en-1-yl}phosphonate (31): Prepared according to the general procedure A. Flash Chromatography: EtOAc (5 % v/v NH_3 aq.) 100; yellow oil, 61 mg, 47 % yield; conv. 91 %; R_f = 0.10 [EtOAc (5 % v/v NH_3 aq.)]; ^{1}H NMR (360 MHz, $CDCl_3$) δ = 6.87 (s, 1H), 6.80 (d, *J* = 0.8 Hz, 2H), 5.14–5.07 (m, 1H), 4.21–4.06 (m, 4H), 3.88 (s, 3H), 3.87 (s, 3H), 3.85 (d, *J* = 9.2 Hz, 1H), 3.68 (dd, *J* = 17.7, 10.0 Hz, 1H), 3.62 (d, *J* = 13.3 Hz, 1H), 1.81 (dd, *J* = 4.9, 1.0 Hz, 3H), 1.58 (dd, *J* = 3.6, 1.2 Hz, 3H), 1.32 (t, *J* = 7.1 Hz, 3H), 1.30 (t, *J* = 7.2 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ = 149.0, 148.1, 138.6 (d, *J* = 13.7 Hz), 132.4, 120.5, 119.9 (d, *J* = 6.1 Hz), 111.5, 110.9, 62.6 (d, *J* = 7.0 Hz), 56.0, 55.9, 53.0 (d, *J* = 158.4 Hz), 50.9 (d, *J* = 17.0 Hz), 26.1, 18.7, 16.6; ^{31}P NMR (101.25 MHz, $CDCl_3$) δ = 25.5; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{18}H_{31}NO_5P$: 372.1934, found 372.1923.

Diethyl 1-(2-Methoxybenzylamino)-3-methylbut-2-enylphosphonate (32): Prepared according to the general procedure A. Flash Chromatography: EtOAc(5wt.-% NH_3)/PE 60:40; yellow oil, 39 mg, 41 % yield, conv. 72 %; R_f = 0.44 (EtOAc = 100); ^{1}H NMR (360 MHz, $CDCl_3$) δ = 7.34–7.29 (m, 2H), 7.14 (dd, *J* = 8.5, 0.7 Hz, 2H), 5.12–5.04 (m, 1H), 4.21–4.05 (m, 4H), 3.88 (d, *J* = 13.6 Hz, 1H), 3.72–3.59 (m, 2H), 2.02 (br. s, 1H), 1.79 (dd, *J* = 4.7, 1.2 Hz, 3H), 1.56 (dd, *J* = 3.6, 1.2 Hz, 3H), 1.31 (t, *J* = 7.1 Hz, 3H), 1.29 (t, *J* = 7.1 Hz, 3H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 148.2, 138.7 (d, *J* = 13.7 Hz), 138.5, 129.5, 120.8, 119.6 (d, *J* = 6.2 Hz), 62.5 (d, *J* = 5.9 Hz), 53.3 (d, *J* = 158.8 Hz), 50.3 (d, *J* = 16.9 Hz), 25.9, 18.5, 16.4; ^{31}P NMR (121 MHz, $CDCl_3$) δ = 25.1; ^{19}F NMR (235.4 MHz, $CDCl_3$) δ = 57.9; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{17}H_{26}F_3NO_4P$: 396.1546, found 396.1536.

Diethyl 3-Methyl-1-(phenylamino)but-2-enylphosphonate (33): Prepared according to the general procedure A. Flash Chromatography: EtOAc/PE = 50:50; yellow oil, 38 mg, 78 % yield; R_f = 0.69 (EtOAc = 100); ^{1}H NMR (300 MHz, $CDCl_3$) δ = 7.23–7.13 (m, 2H), 6.79–6.71 (m, 1H), 6.64 (dd, *J* = 8.5, 0.9 Hz, 2H), 5.16–5.08 (m, 1H), 4.45 (dd, *J* = 20.8, 9.5 Hz, 1H), 4.25–4.02 (m, 5H), 1.81 (dd, *J* = 3.7, 1.2 Hz, 3H), 1.77 (dd, *J* = 4.8, 1.2 Hz, 3H), 1.33 (t, *J* = 7.2 Hz, 3H), 1.29 (t, *J* = 7.2 Hz, 3H);^{13}C NMR (63 MHz, $CDCl_3$) δ = 146.8 (d, *J* = 13.1 Hz), 138.1 (d, *J* = 12.9 Hz), 129.1, 120.0, 118.3, 113.8, 63.0 (d, *J* = 6.9 Hz), 62.7 (d, *J* = 7.0 Hz), 50.7 (d, *J* = 158.9 Hz), 25.8, 18.7, 16.4 (d, *J* = 5.4 Hz); ^{31}P NMR (121 MHz, $CDCl_3$) δ = 24.2; HRMS (ESI) *m/z* [MNa^+] calcd. for $C_{15}H_{24}NNaO_3P$: 320.1386, found 320.1379.

Diethyl [1-({2-[(*tert*-Butyldimethylsilyl)oxy]ethyl}amino)-3-methylbut-2-en-1-yl]phosphonate (34): Prepared according to the general procedure A. Flash Chromatography: EtOAc/PE = 50:50; yellow oil, 56 mg, 53 % yield; R_f = 0.33 (MeOH/CH_2Cl_2 = 5:95); ^{1}H NMR (360 MHz, $CDCl_3$) δ = 5.10–4.98 (m, 1H), 4.19–4.00 (m, 4H), 3.79–3.58 (m, 3H), 2.75 (ddd, *J* = 10.8, 5.9, 4.9 Hz, 1H), 2.59 (dt, *J* = 5.9, 5.3 Hz, 1H), 1.88 (br. s, 1H), 1.76 (dd, *J* = 4.9, 1.1 Hz, 3H), 1.68 (dd, *J* = 3.7, 1.2 Hz, 3H), 1.36–1.21 (m, 6H), 0.87 (s, 9H), 0.02 (s, 6H); ^{13}C NMR (91 MHz, $CDCl_3$) δ = 138.1 (d, *J* = 13.8 Hz), 119.9 (d, *J* = 7.0 Hz), 62.4 (d, *J* = 6.3 Hz), 62.3 (d, *J* = 6.3 Hz), 62.1, 54.4 (d, *J* = 158.1 Hz), 49.7 (d, *J* = 15.9 Hz), 26.0, 25.9, 18.6, 18.2, 16.5, –5.4. ^{31}P NMR (101 MHz, $CDCl_3$) δ = 25.1; HRMS (ESI) *m/z* [MH^+] calcd. for $C_{17}H_{39}NO_4PSi$: 380.2380, found 380.2365.

Acknowledgments

P. A. is the beneficiary of an Allocation Spécifique pour Normalien doctoral research grant from the French Ministry of Higher Education and Research. The authors wish to thank Mr Yann Arribard for technical assistance. We thank the University Paris-Saclay and the CNRS for financial support.

Keywords: Allenes · Phosphonates · Partial reduction · Electrosynthesis

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Received: April 19, 2020

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